

Due April 4, 5:00 PM PST

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Modify this template file to create your submission.

Problem 1a

K-means clustering with 3 clusters of sizes 53, 50, 47

```
Clustering vector:
 [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[38] 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 1 3 1 3 1 3 1 1 1 1 1 3 1 1 1 1 1 1
[75] 3 3 3 3 1 1 1 1 1 1 1 1 1 3 1 1 1 1 1 1 1 1 1 1 1 3 1 3 3 3 3 1 3 3 3 3
[112] 3 3 1 1 3 3 3 3 1 3 1 3 1 3 3 1 1 3 3 3 3 3 1 1 3 3 3 1 3 3 3 1 3 3 3 1 3
[149] 3 1
```

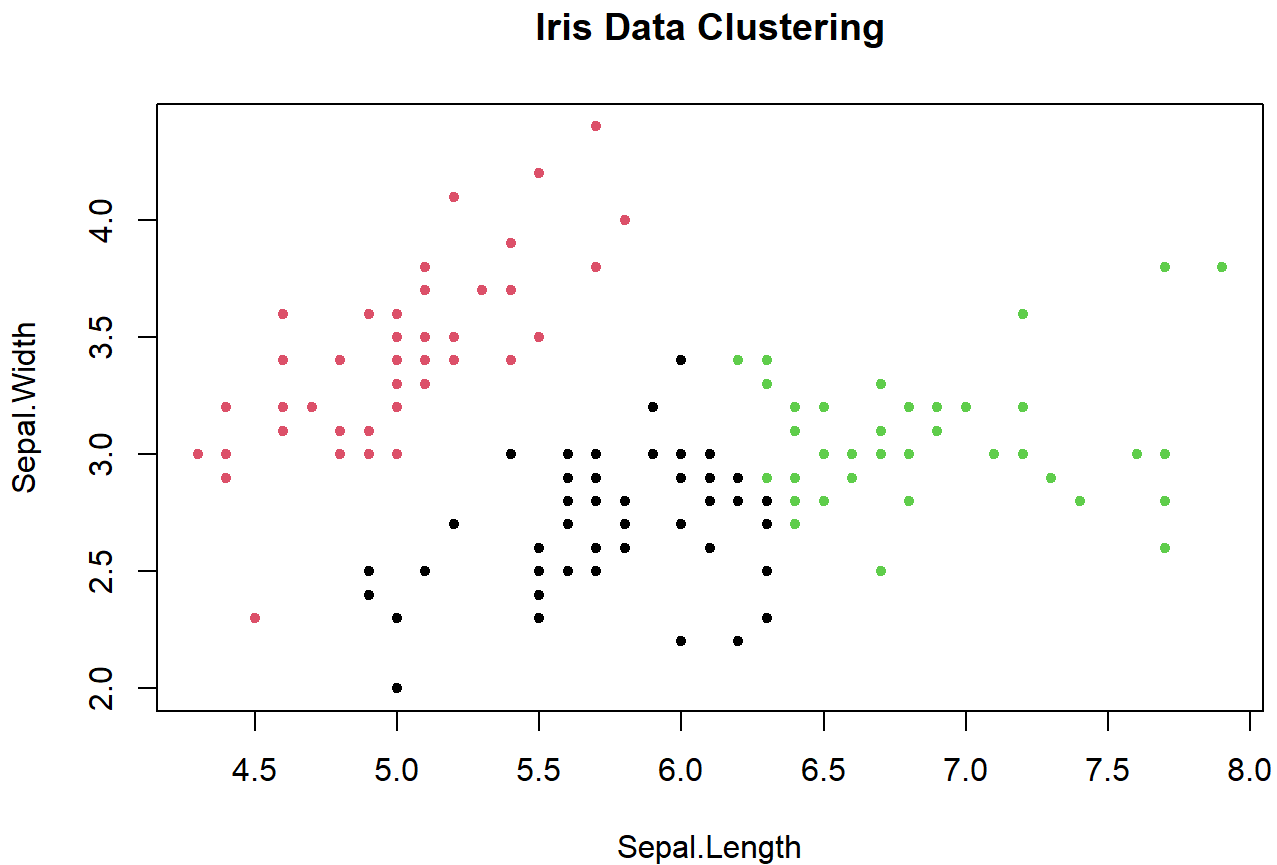
```
Within cluster sum of squares by cluster:
[1] 11.3000 13.1290 12.6217
(between_SS / total_SS = 71.6 %)
```

Available components:				
[1]	"cluster"	"centers"	"totss"	"withinss"
[6]	"betweenss"	"size"	"iter"	"ifault"

Problem 1b

```
## write answer here
```

```
plot(iris_data, col = iris_kmeans$cluster, pch = 20,  
     main = "Iris Data Clustering", xlab = "Sepal.Length", ylab = "Sepal.Width")
```



Problem 2a

```
## write answer here
```

```
library(stringr)
small_sentences <- sentences[1:10]
words_vector <- unlist(str_split(small_sentences, "\\s+"))[1:20]

four_letter_words <- ifelse(str_detect(words_vector, "[A-Za-z]{4}$"), words_vector, NA)

four_letter_words
```

```
[1] NA      NA      NA      "slid" NA      NA      NA      NA      "Glue" NA
[11] NA      NA      NA      "dark" "blue" NA      NA      "easy" NA      "tell"
```

Problem 2b

```
## write answer here
words_bc <- str_subset(words_vector, "^[bcBC]")
words_bc
```

```
[1] "birch"      "canoe"      "blue"      "background."
```

Problem 2c

```
## write answer here
non_alpha_words <- str_subset(words_vector, "^[^A-Za-z]")
non_alpha_words
```

```
[1] "planks."    "background." "It's"
```

Problem 3a

```
## write answer here
pois_llh <- function(theta, n, x) {
  -n * theta + log(theta) * sum(x)
}

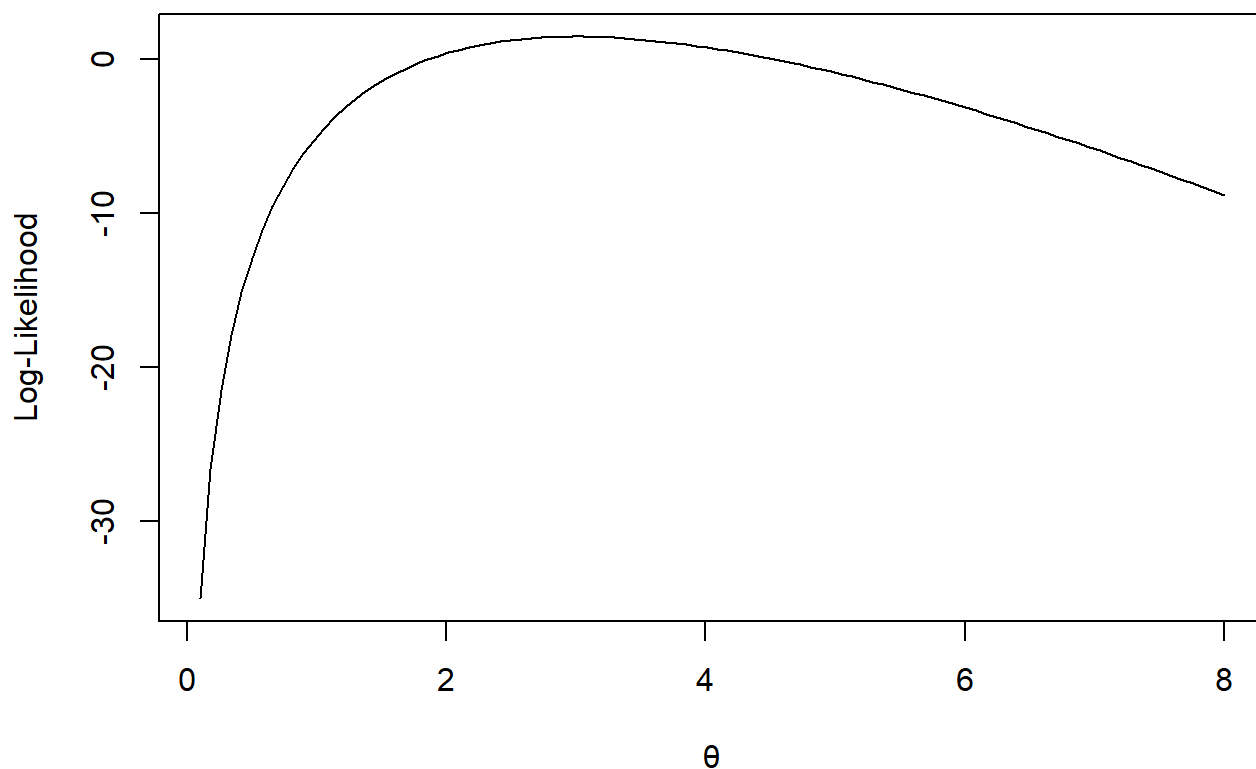
llh_value <- pois_llh(theta = 1, n = 5, x = c(1, 2, 3, 4, 5))
llh_value
```

```
[1] -5
```

Problem 3b

```
## write answer here
theta_vals = seq(from = 0.1, to = 8, length.out = 100)
log_likelihood <- sapply(theta_vals, pois_llh, n = 5, x = c(1, 2, 3, 4, 5))

plot(theta_vals, log_likelihood, type = "l", xlab = "θ", ylab = "Log-Likelihood")
```



```
print("It seems from the plot that the maximum likelihood estimate is around theta = 3.")
```

```
[1] "It seems from the plot that the maximum likelihood estimate is around theta = 3."
```

Problem 4a

```
## write answer here
library(class)

ad_data <- read.csv("ad_data.csv")

set.seed(100)
prop_train <- 0.7
prop_test <- 1 - prop_train
n_data <- nrow(ad_data)
train_test <- sample(c("Train", "Test"),
  size = n_data, replace = TRUE,
  prob = c(prop_train, prop_test))
training_data <- ad_data[train_test == "Train", ]
testing_data <- ad_data[train_test == "Test", ]
```

```
train_pred <- data.frame(EstimatedSalary = training_data$EstimatedSalary)
test_pred <- data.frame(EstimatedSalary = testing_data$EstimatedSalary)
fit_knn <- knn(train = train_pred, test = test_pred, cl = training_data$Purchased, k = 5)
```

Problem 4b

```
## write answer here
accuracy <- mean(fit_knn == testing_data$Purchased)
accuracy
```

```
[1] 0.7777778
```